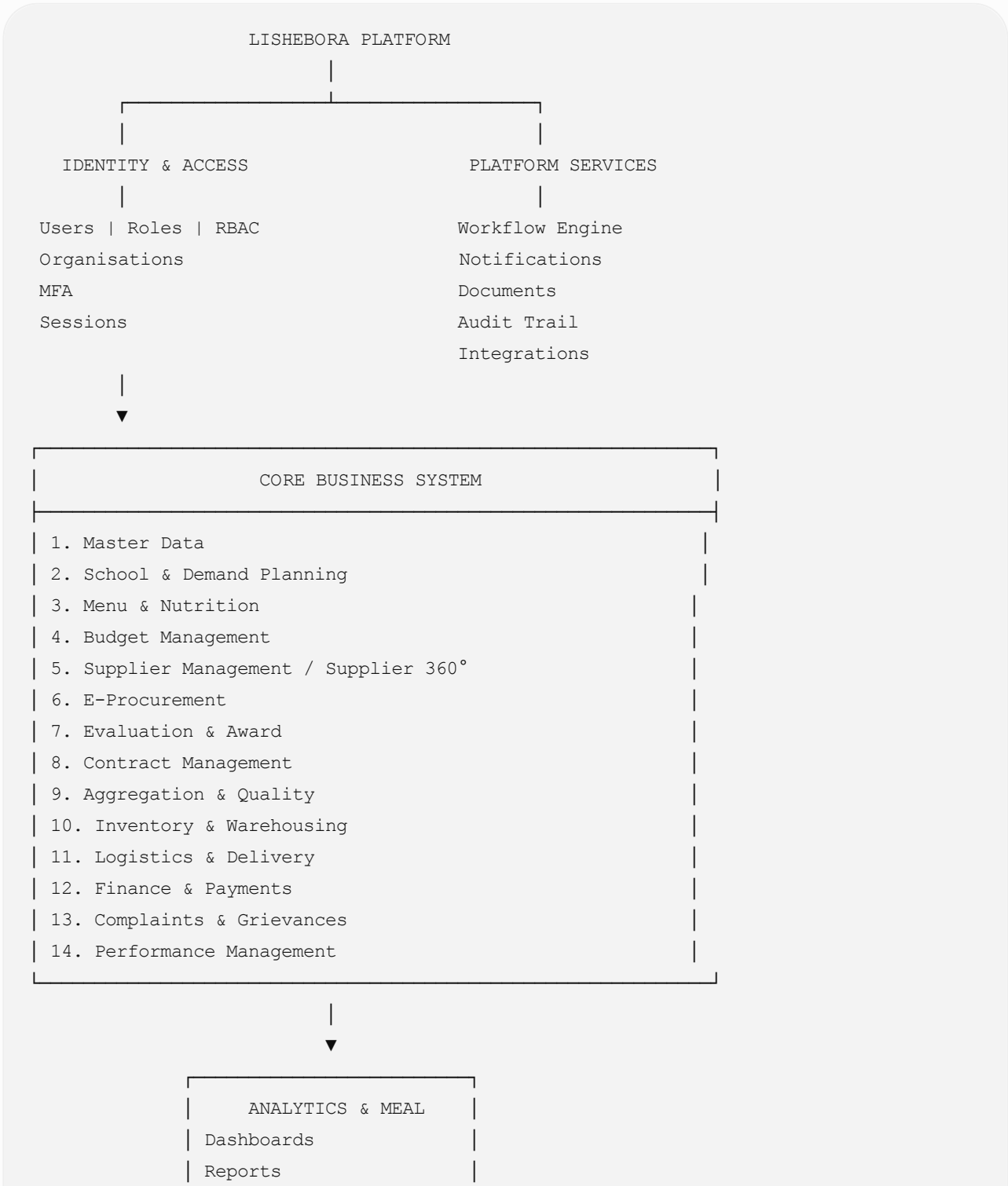


Yes. **With the proposed features document, I would go one step further than my previous structure.** The proposed features make the system much more complete and change it from a basic e-procurement platform into an **integrated school food supply-chain management system with e-procurement at the centre.** The review specifically recommends expanding the original brief with inventory, contracts, budget controls, workflow, supplier 360°, risk, QR traceability, offline synchronization, GIS, notifications, document management and advanced analytics.

LisheBora_System_Document_Revie...

I would structure the final system like this.

LisheBora — Proposed Final System Structure



Forecasting
GIS
Risk & Alerts

The proposed document itself recommends **14 business domains**, including Identity & Access, Master Data, Demand & Nutrition, Budget, Supplier Management, Procurement, Contracts, Aggregation & Quality, Inventory, Logistics, Finance, Complaints/Performance, MEAL and Platform Services.

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1. Identity & Access Management

This should be the foundation.

Features

- Login/logout
- Password management
- MFA
- User registration
- User activation/deactivation
- Role management
- Permission management
- Organisation management
- County/sub-county/school assignment
- User scope
- Session management
- Account locking
- Password reset
- Login history

Important

Permissions should not simply be:

Admin / User

They should be granular:

VIEW
CREATE
EDIT
SUBMIT
VERIFY
EVALUATE

APPROVE
REJECT
PAY
EXPORT
AUDIT

The proposed document specifically recommends this granular permission model.

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2. Master Data Management

This is one of the most important modules because almost everything else depends on it.

Manage

- Counties
- Sub-counties
- Wards
- Schools
- School clusters
- Academic years
- Terms
- Feeding days
- Commodities
- Commodity categories
- Units
- Quality grades
- Nutrition standards
- Food groups
- Portion sizes
- Procurement methods
- Financial thresholds
- Approval thresholds
- Funding sources
- Supplier categories

Design principle

Administrators should configure these without changing source code.

3. School & Demand Management

This becomes the starting point of the procurement cycle.

School enters

- Enrolment
- Expected attendance
- Feeding days
- Menu
- Existing stock
- Opening balance
- Storage capacity
- Budget
- Funding source
- Required commodities
- Required quantities
- Required delivery date

System calculates

```
Students
x
Feeding Days
x
Portion Size
x
Menu
-
Available Stock
=
Required Quantity
```

Then:

```
School Demand
    ↓
School Cluster
    ↓
Sub-county
    ↓
County
    ↓
Consolidated Procurement Requirement
```

The proposed architecture explicitly places demand calculation, consolidation and nutrition indicators in this domain.

4. Budget Management

I would definitely add this as a **standalone module**.

Features

- Create budget
- Funding sources
- Allocate budget to county
- Allocate to school
- Allocate by commodity
- Budget periods
- Approved budget
- Committed amount
- Actual expenditure
- Available balance
- Budget utilisation
- Budget alerts
- Over-budget warning
- Exception approval

Example

School A Budget
KSh 10,000,000

Approved	10M
Committed	7M
Paid	5M
Available	3M

Before issuing a PO:

PO Value = KSh 3.5M

Available Budget = KSh 3M

↓

BLOCK

"Insufficient budget"

This is one of the important gaps identified in the review.

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5. Supplier Management — Supplier 360°

This should be more than supplier registration.

Supplier 360° profile

SUPPLIER

- |
- └ Basic Information
- └ Ownership
- └ Gender/Youth/PWD
- └ Location
- └ Commodities
- └ Capacity
- └ Documents
- └ Certifications
- └ Bids
- └ Awards
- └ Contracts
- └ Purchase Orders
- └ Deliveries
- └ Quality History
- └ Invoices
- └ Payments
- └ Complaints
- └ Performance Score
- └ Risk Status

This is a very good proposed enhancement because management can see the supplier's **complete history in one place.**

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6. E-Procurement

This remains the heart of the system.

Procurement types

- RFQ
- Competitive bidding
- Framework agreement
- Direct sourcing
- Call-off order

Features

- Procurement plan
- Sourcing event

- Lots
 - Specifications
 - Eligibility criteria
 - Evaluation criteria
 - Publication
 - Supplier notification
 - Clarifications
 - Amendments
 - Bid submission
 - Bid locking
 - Bid opening
 - Evaluation
 - Award
 - Purchase order
-

7. Bid Evaluation & Approval

This needs a strong workflow.

```
BID CLOSED
  ↓
TECHNICAL EVALUATION
  ↓
FINANCIAL EVALUATION
  ↓
CONSOLIDATED SCORE
  ↓
RECOMMENDATION
  ↓
APPROVAL
  ↓
AWARD
```

Evaluator features

- Assigned evaluations
- Conflict-of-interest declaration
- Technical score
- Financial score
- Comments
- Submit evaluation

Approval features

- Approve
 - Reject
 - Return
 - Request clarification
 - Escalate
-

8. Contract Management

This is a major improvement over the original system.

Features

- Contract creation
- Contract approval
- Contract versioning
- Contract value
- Contract quantity
- Start/end date
- Delivery schedule
- Call-off orders
- Remaining quantity
- Remaining value
- Contract amendments
- Contract renewal
- Contract expiry alerts
- Supplier contract history
- Contract performance

For example:

Framework Contract

Value: KSh 50M

Used: KSh 32M

Balance: KSh 18M

Quantity:

Contracted: 100,000 kg

Delivered: 65,000 kg

Balance: 35,000 kg

The proposed document specifically identifies contract balances, call-offs, amendments and expiry as missing functionality.

9. Aggregation & Quality

This connects suppliers/farmers to actual food delivered.

Aggregation

- Farmer intake
- Farmer identity
- Commodity
- Variety
- Quantity
- Date
- Location
- Batch number
- Aggregation
- Grading
- Allocation

Quality

- Inspection
- Test results
- Photos
- Accepted quantity
- Rejected quantity
- Reason for rejection
- Corrective action
- Clearance

10. Inventory & Warehouse Management

This should be a full module rather than just a stock field.

Warehouse

- Warehouse registration
- Location
- Capacity
- Manager
- Commodity types

Inventory

- Opening stock
- Receipts
- Issues
- Transfers
- Reservations
- Allocations
- Adjustments
- Damaged stock
- Expired stock
- Rejected stock
- Wasted stock
- Closing stock
- Stock count
- Reconciliation

Alerts

LOW STOCK
EXPIRING STOCK
EXPIRED STOCK
UNUSUAL STOCK LOSS
STOCK VARIANCE

The review specifically proposes these inventory capabilities.

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11. Logistics & Delivery

Features

- Delivery planning
- Route planning
- Vehicle
- Driver
- Dispatch
- Dispatch note
- Delivery status
- GPS where available
- Delivery milestones
- Proof of delivery
- Quantity received

- Quality received
- Shortage
- Damage
- Rejection
- Replacement

Status

READY

↓

DISPATCHED

↓

IN TRANSIT

↓

DELIVERED

↓

ACCEPTED

or

DELIVERED

↓

PARTIALLY ACCEPTED

↓

REPLACEMENT REQUIRED

12. Finance

Invoice

PO

+

Goods Received

+

Invoice

↓

THREE-WAY MATCH

↓

Finance Approval

↓

Payment

Features

- Invoice creation

- Invoice submission
 - Invoice verification
 - Three-way matching
 - Deductions
 - Partial invoices
 - Payment approval
 - Payment reference
 - Payment status
 - Reconciliation
 - Outstanding payments
 - Payment ageing
-

13. Risk & Exception Management

I would add this as a proper dashboard rather than hiding risk inside other modules.

Automatically flag

- Repeated late deliveries
- Supplier quality failures
- Expired certification
- Budget overrun
- Unusual pricing
- Repeated rejected goods
- Large delivery shortages
- Unpaid invoices
- Procurement delays
- Repeated complaints

Then:

```
Risk Detected
  ↓
Risk Assigned
  ↓
Action Required
  ↓
Corrective Action
  ↓
Resolution
  ↓
Closed
```

The proposed document identifies supplier risk, procurement risk, food-safety risk and exception workflows as a missing capability.

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14. QR / Barcode Traceability

I like this proposed feature, but I would put it in **Phase 2**, not the initial MVP.

Example:

BATCH QR CODE

↓

Scan at aggregation

↓

Scan at warehouse

↓

Scan at dispatch

↓

Scan at school

↓

Trace back to supplier

↓

Trace back to farmer group

This would allow:

"Where did this rice delivered to School X come from?"

The review proposes unique batch IDs, QR/barcodes, scanning at aggregation, dispatch and receipt, and recall support.

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15. Offline Functionality

This is particularly important if the platform will operate in rural areas.

The system should allow selected users to work without internet.

Example

Quality Inspector:

Internet available

↓

Download assignment

```

      ↓
GO TO WAREHOUSE
      ↓
No Internet
      ↓
Conduct inspection
      ↓
Take photographs
      ↓
Save locally
      ↓
Internet returns
      ↓
AUTO SYNC

```

Need

- Offline storage
- Queued transactions
- Sync
- Sync status
- Conflict detection
- Conflict resolution
- Retry
- Offline photos

These requirements are explicitly proposed in the review.

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16. GIS

I would have a **map dashboard**.

Map layers

- Schools
- Suppliers
- Farmer groups
- Aggregators
- Warehouses
- Delivery locations

Analysis

School → Supplier distance

School → Nearest aggregator

Commodity supply gaps

Supplier concentration

Delivery routes

County procurement distribution

The review proposes GIS for schools, suppliers, hubs and warehouses, including distance, routes and supply gaps.

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17. Notification Centre

Instead of building SMS/email separately into every module, make it a **central platform service**.

EVENT

↓

Notification Engine

↓

SMS

EMAIL

IN-APP

PUSH

Events

- New RFQ
- Bid deadline
- Approval required
- Contract expiring
- Certificate expiring
- Delivery scheduled
- Delivery delayed
- Invoice approved
- Payment made
- Complaint updated

This is exactly the direction proposed in the review.

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18. Document Management

Central document repository:

Supplier Documents
Contracts
RFQs
Bids
Evaluation Documents
Purchase Orders
Inspection Reports
Delivery Notes
Invoices
Payment Records
Complaints

Each document should have:

- Type
- Owner
- Version
- Expiry
- Access level
- Related transaction
- Uploaded by
- Upload date
- Audit trail
- Retention period

19. Complaints & Grievance Management

Instead of simply having a "complaints" form, make it a **case management system**.

Complaint Submitted
↓
Case Number Generated
↓
Assigned Officer
↓
Investigation
↓
Action
↓
Response
↓
Resolution
↓
Complainant Feedback
↓
Closed

This allows management to identify recurring issues.

20. Supplier Performance

After every completed order:

Supplier Scorecard

Quality	92%
On-time delivery	85%
Completeness	96%
Responsiveness	90%
Compliance	98%

Overall Performance

Good / Needs Review

I would **avoid making this an unexplained single score**. Keep the underlying indicators visible so procurement teams can understand why performance changed.

21. MEAL & Analytics

This should be the management intelligence layer.

Dashboard 1 — Procurement

- Procurement value

- Number of RFQs
- Procurement cycle time
- Award rate

Dashboard 2 — Supplier

- Registered suppliers
- Active suppliers
- Women-led
- Youth-led
- PWD-led
- Supplier performance

Dashboard 3 — Food

- Commodity demand
- Local sourcing
- Rejected food
- Quality failures

Dashboard 4 — Finance

- Budget
- Commitments
- Expenditure
- Outstanding invoices
- Payment time

Dashboard 5 — Programme

- Schools covered
- Farmers reached
- Counties
- Procurement value
- Delivery performance
- Grievances

22. Advanced Analytics — Later

The proposed features are also good here, but I would **not make AI the foundation of the MVP**.

Later:

Demand forecasting

Based on historical enrolment, feeding days, menus and consumption.

Supply forecasting

Expected supplier/producer availability.

Price intelligence

Track commodity price movements.

Anomaly detection

Identify unusual procurement or pricing patterns.

Predictive alerts

"Supplier X has increasing late-delivery risk."

Forecast accuracy

Compare predicted vs actual demand.

These are proposed as future capabilities.

LisheBora_System_Document_Review...

23. Recommended final user roles

With the new features, I would use the following roles:

#	Role
1	Super Administrator
2	System Administrator
3	County Administrator
4	County Procurement Officer
5	County Finance Officer
6	School Administrator
7	School Meals Officer

#	Role
8	Nutrition Officer
9	Procurement Officer
10	Evaluation Committee Member
11	Approving Officer
12	Supplier
13	Farmer Group / Cooperative
14	Aggregator
15	Warehouse Officer
16	Quality Inspector
17	Logistics Officer
18	Driver
19	Finance Officer
20	Programme/MEAL Officer
21	Risk/Compliance Officer
22	Auditor
23	Helpdesk Officer

Some of these can be combined during the pilot if the organisation has limited staff. The proposed review already identifies most of the core roles and recommends granular permissions rather than broad user categories.

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24. The final architecture I would recommend

The system should ultimately look like this:



The proposed review's final conclusion describes essentially this same direction: an integrated school food supply-chain platform with **e-procurement at its core**, moving through demand, budget, sourcing, evaluation, contracts, aggregation, quality, inventory,

logistics, delivery, invoice, payment, performance and MEAL.

LisheBora_System_Document_Revie...

What I would change from my previous answer

The **previous structure was good for the original brief**, but with this new proposed-features document, I would now make these **first-class modules**:

Budget Management → Contract Management → Inventory/Warehouse → Risk & Exceptions → Supplier 360° → Workflow Engine → Document Management → Notification Engine → Traceability → GIS

And I would keep **AI/forecasting, full offline synchronization, advanced GIS, QR traceability and deep external integrations for later phases**, rather than making them mandatory for the first release. The proposed document itself recommends this phased approach.

LisheBora_System_Document_Revie...

So, before development, the next document I would create is a proper *LisheBora Functional Requirements & Role-Permission Matrix*. That document should go module by module and specify **every feature, screen, field, action, user role, approval, status, notification, validation rule and report**. That is the level at which you can confidently hand the specification to a developer without leaving major decisions for the developer to guess.

Sources